Problem · Topic 3 · 4 points · 60 min

P15 — CAESA: financing a second autoclave with a semiannual loan or with new shares

The chief financial officer (CFO) of ***COMPOSITES AEROESPACIALES DE ÁLAVA S.A. (CAESA)****, a company of Vitoria dedicated to the manufacturing of carbon-fibre fairings for helicopters, is closing the books of the year 2021. The balance of the company at 31 December 2020 (31/12/2020) is included below; the operations of the period from 1 January 2021 (01/01/2021) to 31 December 2021 (31/12/2021) have been the following:*

I. *The company has sold 1.200 fairings at a price of 30 m.u. / unit (every month the same amount, 100 u.). Of the sales of every month, 20% is charged in cash and the remaining 80% is payable to 60 days. The customer debt of the previous year was charged in January, and the suppliers and the unpaid taxes of the previous year were paid in January and in April respectively.*

II. *At 31 December 2020 (31/12/2020) there is a stock (stock in stock) of 500 kg of epoxy resin valued at 3 m.u. / kg. On 1 March 2021 (01/03/2021) the company purchased 4.000 kg at 3 m.u. / kg, paying 30% in cash and the remaining 70% in two equal parts, on 1 September 2021 (01/09/2021) and on 1 March 2022 (01/03/2022). At 31 December 2021 (31/12/2021) there remain 700 kg in the warehouse.*

III. *The fixed salaries are 1.000 m.u. / month, paid in the same month, to which 300 m.u. / month of Social Security must be added, paid the month after. Variable wages of the lamination staff amounted in 2021 to 1.200 m.u. (paid in cash). The external supplies (energy for the autoclaves) are 250 m.u. / month and are paid two months after being made.*

IV. *The autoclave of the balance was bought on 1 January 2018 (01/01/2018) and is amortized linearly in 10 years with a residual value of 1.000 m.u. On 1 July 2021 (01/07/2021) CAESA put into service a second autoclave of 6.000 m.u. (amortized linearly in 10 years with a residual value of 600 m.u.) together with a software license for the control of the curing cycles of 1.200 m.u. (amortized linearly in 3 years with residual value zero), both paid in full in cash that day.*

V. *To pay the 7.200 m.u. of the expansion, the board accepted on 1 July 2021 (01/07/2021) a bank loan of 7.200 m.u. at 5% annual interest, to be returned in 8 semiannual installments of 900 m.u. every 1 January and every 1 July, the first on 1 January 2022 (01/01/2022); the same days the company pays the interest of the semester calculated on the debt at the beginning of the semester. The long-term loan of the balance bears 6% annual interest, paid every 31 December, and is returned in one single payment on 31 December 2023 (31/12/2023).*

VI. *The financial results are declared at 23:59 hrs on 31 December 2021 (31/12/2021). The corporation tax is 20% on the Profits and is paid immediately. Of the profit after taxes, 10% goes to the legal reserve, 30% is distributed as dividends paid in cash at 24:00 on 31 December 2021 (31/12/2021), and the rest goes to voluntary reserves.*

VII. *The alternative rejected by the board was a capital increase on 1 July 2021 (01/07/2021) of 4.800 new shares of 1 m.u. nominal value with an issue premium of 50% over the nominal value, fully disbursed in cash, instead of the bank loan.*

BALANCE 31 December 2020 (31/12/2020)

ASSETS	*m.u.*	*LIABILITIES*	*m.u.*
NON-CURRENT ASSETS	***9.600***	***OWNERS EQUITY***	***12.000***
Land	*3.000*	*Paid-in capital (10.000 shares × 1 m.u.)*	*10.000*
Autoclave	*9.000*	*Reserves*	*2.000*
Accum. depreciation	*(2.400)*	***NON-CURRENT LIABILITIES***	***1.000***
CURRENT ASSETS	***5.000***	*Long-term loan (31/12/2023)*	*1.000*
Stock of resin	*1.500*	***CURRENT LIABILITIES***	***1.600***
Accounts receivable	*2.000*	*Suppliers*	*1.200*
Cash	*1.500*	*Unpaid taxes*	*400*
Total Assets	***14.600***	***Total Liabilities + OE***	***14.600***

Required:

1. *Construct the Income Statement (Profit and Loss Account) of CAESA for the year 2021, justifying the depreciation of the old autoclave from the data of the balance.*
2. *Construct and justify the balance sheet at 24:00 on 31 December 2021 (31/12/2021), detailing the cash calculations and the classification of the new loan between current and non-current liabilities.*
3. *Determine the ROA, the ROE and the debt-to-equity ratio of 2021.*
4. *Indicate what changes would have occurred in the Income Statement, the cash and the balance sheet if the board had chosen the capital increase described in VII. Compare the ROE and the debt-to-equity ratio of both alternatives and explain, reasoned, the effect of the financial leverage.*

Setup

Hypotheses:

1. "Payable to 60 days" leaves the November and December sales uncollected at 31/12/2021.
2. The interest of the new loan for July–December 2021 is accrued in 2021 and paid on 1 January 2022, the day after the closing; the installment of that day and the one of 1 July 2022 are due within one year.
3. Land is not depreciated.

Financing sources involved: a long-term bank loan (external liability, [Slide 31]) against a rights issue with premium (external equity, [Slides 25-26]); the reserves are self-financing [Slide 23]. The debt-to-equity ratio is course material [Slide 36].

Extension (not defined in the course decks). $\mathrm{ROA} = \text{Income from operations}/\mathrm{TA}$; $\mathrm{ROE} = \text{Net profit}/E$, on the balance at 24:00.

Part (a) · 1 point

Prepare the 2021 Income Statement

Step 1. Depreciation

$$D_{\text{old}} = \frac{9.000 - 1.000}{10} = 800 \quad (2018\text{–}2020\text{: } 3 \times 800 = 2.400 \checkmark)$$

$$D_{\text{new}} = \frac{6.000 - 600}{10} \times \frac{6}{12} = 540 \times \frac{6}{12} = 270, \qquad D_{\text{software}} = \frac{1.200}{3} \times \frac{6}{12} = 200$$

$$D_{2021} = 800 + 270 + 200 = 1.270 \text{ m.u.}$$

Step 2. Resin consumed

$$\text{Consumption} = 1.500 + 4.000 \times 3 - 700 \times 3 = 1.500 + 12.000 - 2.100 = 11.400 \text{ m.u.}$$

Step 3. Interest

$$I_{\text{old loan}} = 0{,}06 \times 1.000 = 60, \qquad I_{\text{new loan}} = 7.200 \times 0{,}05 \times \frac{6}{12} = 180, \qquad I = 240 \text{ m.u.}$$

Step 4. Income Statement 2021

Line	Computation	m.u.
Sales revenue	1.200×30	36.000
– Resin consumed	Step 2	11.400
– Fixed salaries and Social Security	$12 \times (1.000 + 300)$	15.600
– Variable wages		1.200
– External supplies (energy)	12×250	3.000

Line	Computation	m.u.
– Depreciation and amortization	Step 1	1.270
= Income from operations	$36.000 - 32.470$	**3.530**
– Financial expenses	Step 3	240
= Profit before taxes	$3.530 - 240$	**3.290**
– Corporation tax (20%)	$0{,}20 \times 3.290$	658
= Net profit	$3.290 - 658$	**2.632**

KEY RESULTS

- Old autoclave 800 m.u./year ($3 \times 800 = 2.400$ (checked)); new autoclave 270; software 200; total 1.270 m.u.
- Resin consumed 11.400; income from operations 3.530; interest 240 m.u.
- Profit before taxes 3.290; tax 658; net profit 2.632 m.u.

Part (b) · 1.5 points

Prepare the balance sheet at 24:00 on 31 December 2021

Step 1. Distribution

$$\text{Legal reserve} = 0{,}10 \times 2.632 = 263{,}20, \qquad \text{Dividends} = 0{,}30 \times 2.632 = 789{,}60, \qquad \text{Voluntary reserves} = 0{,}60 \times 2.632 = 1.579{,}20$$

Step 2. Cash account 2021

Inflows	m.u.	Outflows	m.u.
Cash sales ($0{,}20 \times 36.000$)	7.200	Suppliers 2020	1.200
Credit sales January–October (10×2.400)	24.000	Unpaid taxes 2020	400
Receivables 2020	2.000	Resin ($0{,}30 \times 12.000 + 0{,}35 \times 12.000$)	7.800
New bank loan	7.200	Fixed salaries	12.000
		Social Security, January–November	3.300
		Variable wages	1.200
		External supplies, January–October	2.500
		Second autoclave and software	7.200
		Interest of the old loan	60
		Corporation tax 2021	658
		Dividends	789,60

Inflows	m.u.	Outflows	m.u.
Total	**40.400**	**Total**	**37.107,60**

$$\text{Cash}_{24:00,\,31/12/2021} = 1.500 + 40.400 - 37.107{,}60 = 4.792{,}40 \text{ m.u.}$$

Step 3. Classification of the new loan

Installments on 01/01/2022 and 01/07/2022: $2 \times 900 = 1.800$ m.u. current; the other six, $6 \times 900 = 5.400$ m.u., non-current.

Step 4. Balance sheet at 24:00 on 31/12/2021

Assets	m.u.	Liabilities and owners equity	m.u.
Land	3.000	Paid-in capital	10.000
Old autoclave (9.000 − 3.200)	5.800	Legal reserve	263,20
New autoclave (6.000 − 270)	5.730	Voluntary reserves (the 2.000 of 2020 + 1.579,20)	3.579,20
Software license (1.200 − 200)	1.000	**Owners equity**	**13.842,40**
Non-current assets	**15.530**	Long-term loan (31/12/2023)	1.000
Stock of resin (700 kg × 3)	2.100	New bank loan, non-current part	5.400
Accounts receivable (2 × 2.400)	4.800	**Non-current liabilities**	**6.400**
Cash	4.792,40	New bank loan, installments in 2022	1.800
		Interest payable (01/01/2022)	180
		Suppliers of resin (01/03/2022)	4.200
		Social Security payable	300
		External supplies payable	500
Current assets	**11.692,40**	**Current liabilities**	**6.980**
Total assets	**27.222,40**	**Total liabilities and equity**	**27.222,40**

KEY RESULTS

- Legal reserve 263,20; dividends 789,60; voluntary reserves 1.579,20 m.u.
- Cash 4.792,40 m.u.
- New loan: 1.800 current and 5.400 non-current; interest payable 180.
- Owners equity 13.842,40; total 27.222,40 m.u.

Part (c) · 0.5 points

Compute ROA, ROE and debt-to-equity of 2021

$$\text{ROA} = \frac{3.530}{27.222{,}40} = 12{,}97\%, \qquad \text{ROE} = \frac{2.632}{13.842{,}40} = 19{,}01\%, \qquad \frac{L}{E} = \frac{13.380}{13.842{,}40} = 0{,}97$$

KEY RESULTS

- ROA = 12,97%; ROE = 19,01% (extension); $L/E = 0{,}97$.

Part (d) · 1 point

The capital increase instead of the loan: changes, ROE, leverage and the leverage effect

Step 1. New results

$$\text{Profit before taxes} = 3.530 - 60 = 3.470, \qquad \text{Tax} = 0{,}20 \times 3.470 = 694, \qquad \text{Net profit} = 3.470 - 694 = 2.776$$

$$\text{Legal reserve} = 277{,}60, \qquad \text{Dividends} = 832{,}80, \qquad \text{Voluntary reserves} = 1.665{,}60$$

Step 2. New cash

The 7.200 m.u. from the shares replace the 7.200 m.u. of the loan; only the tax (+36) and the dividend (+43,20) change:

$$\text{Cash} = 4.792{,}40 - 36 - 43{,}20 = 4.713{,}20 \text{ m.u.}$$

Step 3. Changes in the balance sheet

Item	Loan	Capital increase	Change
Cash	4.792,40	4.713,20	-79,20
Total assets	**27.222,40**	**27.143,20**	**-79,20**
Paid-in capital	10.000	14.800	+4.800

Item	Loan	Capital increase	Change
Issue premium	0	2.400	+2.400
Legal and voluntary reserves	3.842,40	3.943,20	+100,80
Owners equity	**13.842,40**	**21.143,20**	**+7.300,80**
New bank loan (current and non-current)	7.200	0	−7.200
Interest payable	180	0	−180
Total liabilities and equity	**27.222,40**	**27.143,20**	**−79,20**

Step 4. Comparison and leverage effect

Ratio	Loan	Capital increase
ROA	12,97%	$3.530/27.143{,}20 = 13{,}01\%$
ROE	19,01%	$2.776/21.143{,}20 = 13{,}13\%$
L/E [Slide 36]	0,97	$6.000/21.143{,}20 = 0{,}28$

The assets earn the same income from operations in both cases (about 13% of the assets), while the new loan costs only 5%. Each m.u. borrowed earns more than it costs, and the difference goes to the shareholders: with the loan, equity is 7.300,80 m.u. smaller while the net profit is only 144 m.u. lower, so the ROE rises from 13,13% to 19,01%. (Separately, the profit is shared among 10.000 shares instead of 14.800, which raises the earnings per share, an extension.) This is the positive **financial leverage effect** (the relation ROE vs ROA – cost of debt is an extension). Its price is risk: the loan raises the debt-to-equity ratio from 0,28 to 0,97 and commits the company to 1.800 m.u. of installments plus interest every year, while the new shares never have to be repaid [Slides 25, 36].

KEY RESULTS

- Interest 60; profit before taxes 3.470; tax 694; net profit 2.776 m.u. (+144).
- Legal reserve 277,60; dividends 832,80; voluntary 1.665,60; cash 4.713,20 (−79,20).
- Owners equity 21.143,20; liabilities 6.000; total 27.143,20 m.u.
- ROE 13,13% vs 19,01%; L/E 0,28 vs 0,97. The loan raises ROE because ROA ≈ 13% > cost of debt 5%.

Interpretation and checks

- **Dimensional check:** units × m.u./unit and kg × m.u./kg give m.u.; ratios dimensionless.
- **Profit vs cash:** the net profit of 2.632 m.u. does not explain the cash: 7.200 m.u. of new debt came in and 7.200 m.u. went into fixed assets, and 4.800 m.u. of sales are still uncollected.
- **The day after the closing:** the interest of the new loan is an expense of 2021 although it is paid on 01/01/2022; forgetting it overstates the profit by 180 m.u.
- **Typical mistakes:** putting the whole new loan in non-current liabilities, depreciating the new assets twelve months, ignoring the residual values and treating the issue premium as capital.